

BHARATH KUMAR RAJESH

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Engineer who ships production systems and builds the measurement layer that proves they still work. Ran a multi-agent platform on LangGraph and Claude (Amazon Bedrock) at 10,000+ daily users and 99.5% uptime, with the 200-test grader suite that gated every release. Ships open-source evaluation and integration tooling, including agent-triage on PyPI and ForgeSync, reconciliation infrastructure for construction finance.

SKILLS

Evaluation & Testing: Eval Harnesses, Automated Graders, Benchmark Design, Regression Taxonomies, Drift Detection, Mutation Testing, Vitest, Playwright, pytest, TDD

LLM & Agents: LangGraph, Claude (Amazon Bedrock), MCP, RAG, LLM Provider APIs (Streaming/SSE), Prompt Versioning, OpenRouter, LiteLLM, Fireworks

Backend & Services: TypeScript (strict), Fastify, Prisma, BullMQ, Zod, FastAPI, Flask, Spring Boot, REST, OpenAPI, gRPC, SSE, React, Next.js

Languages: Python (advanced), TypeScript, Java, SQL, Go

Cloud & Infra: AWS (EC2, S3, Lambda, ECS, SageMaker), AWS CDK, PostgreSQL, Redis, Docker, Docker Compose, Kubernetes, Terraform, GitHub Actions, CI/CD

Observability: OpenTelemetry, OpenTelemetry GenAI Semconv, OTLP, Grafana, Prometheus, Structured Logging

EXPERIENCE

Applied AI Engineer (Graduate Assistant appointment)

March 2025 to May 2026

Pace University, Seidenberg School of Computer Science

New York, NY

- Built the evaluation pipeline the platform shipped against: **200+ automated grader tests per release** scoring per-prompt quality in **CI**, which caught **4 silent prompt-template regressions** before they reached production.
- Shipped production AI agents on **LangGraph** and **Claude (Amazon Bedrock)** serving **10,000+ daily users** at **99.5% uptime** across **12 zero-regression releases**.
- Bridged live systems of record through **MCP** servers and handled the failure modes that actually break production integrations: malformed upstream responses, silent tool-call drift, and auth expiry mid-session.
- Unified **50,000+ documents** scattered across sources into one retrieval pipeline reaching **92% MRR@5**, cutting retrieval latency **60%**.
- Instrumented the platform with **OpenTelemetry** and **Grafana** to track latency, token throughput, and quality drift, making model behavior measurable instead of anecdotal.
- Ran as day-to-day engineering contact for university stakeholders, turning vague asks into shipped product through weekly reviews with leadership and IT.

Head of Research and Development (early-stage AI startup, pre-incorporation, two-engineer team)

May 2024 to July 2024

Metis

Bangalore, India

- Built retrieval and analysis pipelines for **Rainmaker**, the company's flagship financial analysis and forecasting product, working in Python across **Azure** and **OpenAI** model integrations.
- Shipped changes through reviewed pull requests as one of two engineers on the product repository, owning the research side of the build from notebook prototype to versioned release.

Software Development Intern

September 2023 to May 2024

Let's Be The Change

Bangalore, India

- Cut API **P99 latency 80%, from 2,000ms to 400ms**, by re-architecting query patterns and introducing a **Redis** cache layer.
- Integrated **Stripe**, **SendGrid**, and **Twilio** end to end behind **Java** and **Spring Boot** APIs and webhooks, with zero post-launch incidents across **1,000+ users**.
- Shipped an A/B test on recommendation logic in under 3 weeks, producing a **20% retention lift** and **35% reduction in onboarding abandonment**.

PROJECTS

ForgeSync | TypeScript strict, Fastify, PostgreSQL, Prisma, Redis, BullMQ, React, AWS CDK | github.com/thebharathkumar/ForgeSync

- Reconciliation infrastructure for construction finance. Ingests invoices from **4 connectors** with genuinely different wire formats, normalizes to one canonical model where money is an integer count of minor units, then matches records with **5 weighted, fully decomposable signals** and shows the per-signal breakdown rather than a black-box score.
- Every mutation to an external system requires human approval and is **idempotent at three layers**, with an append-only audit trail of before state, after state and actor. **125 unit and 75 integration tests**, the integration suite against real PostgreSQL and Redis rather than fakes, deployed via **AWS CDK** that synthesizes without credentials.

loopcheck | Python, mutation testing, confidence scoring, HMAC audit chain | github.com/thebharathkumar/loopcheck

- Verifier-first agent loop that grades its own grader: **mutation testing as ground truth**, so a test only counts if it catches deliberately broken code. Reports **precision 1.00 and recall 0.33 at the 0.85 accept threshold** against a 15-file labelled set, publishing the weak recall alongside the strong precision.

agent-triage | Python, FastAPI, OpenTelemetry, OTLP, SSE | github.com/thebharathkumar/agent-triage

- Backend service and dashboard for triaging production multi-agent failures, with an OTLP span receiver, SSE streaming, and LLM root-cause narratives. **220+ tests, 93%+ coverage**, published on **PyPI**.

downgrade | Python, OpenTelemetry GenAI Semconv, OpenRouter, LiteLLM | github.com/thebharathkumar/downgrade

- Auto-evaluation harness for generated text, measuring silent quality regression when a model router downgrades a request across **six routing arms**, using **four structural detectors plus two LLM judges** over a **six-type regression taxonomy**.

Open Source: faramesh-core: 9 PRs including merged PR #33 (delegation token policy). Layer5 / Meshery (CNCF): 5 merged. Artie Labs: opened three Go PRs.

Publications: Springer Nature, ICACECS 2023: deep CNN plant and medicinal species classification, 89% accuracy, corresponding author. IJARESM, November 2023: driver drowsiness detection, Dlib EAR plus Keras CNN, 93% pass rate.

EDUCATION

M.S. Computer Science, GPA 3.87/4.0	August 2024 to May 2026
<i>Pace University, Seidenberg School of Computer Science</i>	<i>New York, NY</i>
B.E. Computer Science	2020 to 2024
<i>Visvesvaraya Technological University</i>	<i>Bangalore, India</i>

Certifications: AWS Solutions Architect Associate (2026). Anthropic Claude with Amazon Bedrock (2026). RHCSA (2022).